Ryerson University
Department of Computer Science
CPS125 - Winter 2016
Midterm Test - Section IX (Programming)

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Please circle your professor's name

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Important Notes:

- This section must be written in pen.
- For all questions requiring reading from a file, you must the file input protocol (fopen/fscanf/fclose).
- You must use scalar variables only. Use of arrays is forbidden.

FOR OFFICE USE ONLY

1 to 20	21	22	TOTAL
9	5	4.5	18.5
/10	/5	/5	/20

Do not open the test until instructed

Professors and invigilators will not answer any questions during the test
(except if you need help understanding an English word)

Question 21 (5 marks):

A file named "numbers.dat" contains 20 integer numbers (no duplicates, all the numbers are different). Write a complete C program that prints out the two smallest numbers.

```
#include <stdio.h>
#include <limits.h>

int main () {

int small1 = INT_MAX, small2 = INT_MAX ;
FILE *in ;
int num ;

in= fopen ("numbers.dat", "r") ;

while (!feof (in)) {
fscanf (in, "%d" ,&num) ;
~~if (small1 ;~~
if (num < small1) {
    small2 = small1 ;
    small1 = num ;
} else if (num < small2) {
    small2 = num ;
}
}
fclose (in) ;
printf ("smallest number is: %d, while second smallest is: %d", small1, small2) ;
return 0 ;
}
```

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Question 22 (5 marks):

Gosper's formula is used to calculate an approximation of the factorial of a number.

$$n! = n^n e^{-n} \sqrt{(2n + 1/3)\pi}$$

Write a complete C program that prompts the user to enter a positive integer number n (you must write an input validation loop to require a positive value), uses Gosper's formula to approximate n!, and then displays the result with three decimals precision.

Note that the values of e and pi are predefined in math.h as constant macros M_E and M_PI.

```
#include <stdio.h>
#include <math.h>

int main () { double factorial ; double n;
printf ("Input a positive integer number\n") ;

do {
printf ("Input a positive integer number \n") ;
scanf ("%d", &n) ;

} while (num <=0) ;

factorial = pow(n,n) * exp(-n) * sqrt((2*n + 1.0/3) * M_PI) ;

printf ("n! is %.3f", factorial) ;

return 0 ;

}
```

0+2+2+0.5
= 4.5

End of programming section.

Don't forget to do the multiple choice part.

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